A Controlled Reasoning-SFT Trajectory Reveals Multi-Timescale Routing Reorganization in Sparse Mixture-of-Experts Models

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Abstract

Sparse mixture-of-experts (MoE) routers map continuous score profiles to discrete top-k expert sets, but endpoint comparisons do not reveal when or at what resolution routing organization changes. We follow one Qwen3-30B-A3B model through a controlled 1,000-step full-parameter reasoning-SFT run, saving 15 checkpoints and teacher-forcing the same 64 traces, comprising 36,853 prompt-plus-continuation tokens, at every state. By step 1,000, expert use is more diffuse (N_{eff}: $66.6 \rightarrow 89.8$; Gini: $.601 \rightarrow .456$), while baseline-relative top-8, rank-window, and shared-ID rank drift reach .453, .763, and .321; the geometric cutoff-margin ratio contracts to .636. These observables do not share one timescale. Step 50 already accounts for 76.6%, 85.0%, and 79.6% of the respective endpoint local drifts, but only 28.2% and 26.7% of the endpoint changes in effective count and Gini. Interval-width-normalized adjacent-checkpoint drift generally declines, with local fluctuations, but remains non-zero over $900 \rightarrow 1000$, while summed adjacent-checkpoint dissimilarity greatly exceeds endpoint dissimilarity. A fixed 128-prompt behavior audit, sampled from the official GSM8K training split and held out from the SFT corpus, is complete through step 900. Under capped greedy decoding, measured exact-match accuracy fluctuates rather than improving monotonically as routing drift accumulates, while generation length and 512-token censoring rise strongly; step 1,000 behavior remains excluded pending completion. Cross-sectional Qwen replay and Kanana, Nemotron, and Gemma4 comparisons support the separability of aggregate use, score rank, margin, rank-window membership, and selected identity, but not a universal direction or magnitude. The tested cross-fitted global-permutation controls do not recover held-out local routing. A separate normalized margin-erosion assay prospectively targets routing transitions and routing restoration suppresses next-token divergence among observed flips, while a balanced pilot does not resolve an interaction between reorganization and vulnerability. The central result is therefore a multi-resolution, multi-timescale training trajectory: substantial local routing reorganization is already present at the earliest observed checkpoint, step 50, while aggregate allocation continues to evolve; neither should be treated as a proxy for behavioral capability or perturbation vulnerability.

1 Introduction

Sparse MoE routers create a discrete expert-selection mechanism: a continuous score profile is converted into a top-k identity set at every token (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022). Post-training can change model behavior substantially, but it is unclear whether it mainly redistributes aggregate expert usage or reorganizes the local competitions that determine individual routing decisions. Aggregate load, entropy, and frequency cannot answer this question: a checkpoint can preserve coarse utilization while changing which experts participate and which experts lie near a routing cutoff.

We ask three questions: *when does routing change during post-training, what organizational resolutions change, and do those changes move in lockstep with behavior on prompts held out from the SFT corpus?* Our primary evidence is a controlled full-parameter reasoning-SFT trajectory of one Qwen model from initialization through step 1,000. At Base and 15 saved checkpoints we replay exactly the same token IDs and measure aggregate load, selected identity, rank-window membership, shared-ID rank agreement, and boundary margin. This design converts post-training from an endpoint label into an observed optimization trajectory while removing generated-text variation from the routing comparison.

The trajectory reveals neither a single scalar shift nor a simple convergence curve. Local identity,

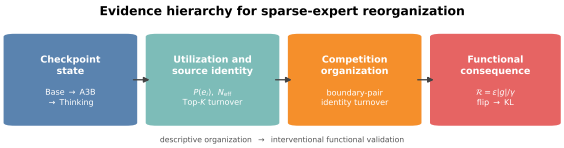


Figure 1: Evidence hierarchy. The controlled same-content trajectory asks when and at what resolution routing changes. Cross-sectional checkpoints and model families test scope. Local interventions separately test the vulnerability and consequence of the resulting routing states.

rank-window, and rank observables undergo most of their net endpoint change by the earliest saved checkpoint, whereas aggregate allocation and margin evolve more slowly. Adjacent-state turnover persists even after baseline-relative drift changes little, so the sampled route through training is much longer than the final displacement from Base. The completed portion of a fixed 128-item behavior audit also does not support a simple narrative in which greater routing drift tracks monotone accuracy improvement: accuracy fluctuates while generation length and truncation increase. Because checkpoint-1,000 behavior is unfinished, we treat this as a bounded result through step 900 rather than a completed endpoint claim.

The remaining experiments test the scope and consequence of this trajectory result. Cross-sectional Qwen Base/A3B/Thinking comparisons and Common-Trace Replay show that local turnover can be hidden by similar coarse utilization. Kanana, Nemotron, and Gemma4 test whether the score–rank–margin–identity decomposition is observable beyond Qwen; they do not define a universal training law. Finally, contribution-space interventions test functional vulnerability as a separate property of a routing state. We do not claim that measured score change causes identity turnover, that one trajectory generalizes to all post-training, or that routing reorganization causes behavioral change.

Controlled temporal evidence is primary; checkpoint and architecture comparisons provide external triangulation; interventions test a distinct functional question. Scaling a realized post-gating contribution by $1 + \delta_i$ and propagating the perturbed hidden state through later routers defines the normalized margin-erosion assay $\mathcal{R} = \epsilon|g|/\gamma$. This assay probes the vulnerability of the resulting routing state and is not a definition of reorganization.

Our contributions are a same-content, 15-checkpoint routing trajectory that characterizes distinct empirical temporal profiles; a resolution-based account, tested against global relabeling and triangulated across three additional MoE families; and bounded evidence that routing reorganization is neither a scalar capability proxy nor the same object as local perturbation vulnerability.

2 Related work

Routing, balance, and stability in sparse MoEs. Sparse gating enables conditional capacity (Shazeer et al., 2017), while GShard, Switch Transformers, BASE Layers, and Hash Layers develop scalable or balancing-oriented routing strategies (Lepikhin et al., 2021; Fedus et al., 2022; Lewis et al., 2021; Roller et al., 2021). ST-MoE studies stable transfer and fine-tuning, and Stable-MoE explicitly targets assignment fluctuation during training (Zoph et al., 2022; Dai et al., 2022). Auxiliary-loss-free balancing further shows that utilization objectives and routing optimization need not be reduced to one formulation (Wang et al., 2024). These works make routing dynamics and load control established concerns; our novelty is not the generic claim that routing can change. We instead observe one fixed-seed reasoning-SFT run at 15 intermediate states and compare the temporal profiles of aggregate load, selected identity, rank-window membership, shared-ID rank, and margin under identical teacher-forced content.

Expert specialization and routing interpretation. Expert specialization is neither guaranteed nor necessarily captured by a single semantic label (Krishnamurthy et al., 2023). Recent routing analyses examine task-conditioned activation signatures and question simple domain-specialist accounts (Avinash, 2026; Wang et al., 2026; Lasy et al., 2026). We address a complementary axis: stability across training states at aligned token sites. The tuple $\mathcal{C}(x, \ell, t)$ is not a new similarity metric; it represents local competition. Aggregate utilization is a separate marginal statistic, while Jaccard, shared-ID rank agreement, and margin ratios compare local objects across states. Their distinct trajectories show why none of these resolutions substitutes for all the others. Same-content replay and cross-fitted permutation controls additionally distinguish local reorganization from generated-text change and show that the tested global relabelings fail to recover held-out local routes. The methodological

contribution is this organizational resolution, not the individual similarity statistics used to measure it.

Fine-tuning and functional interventions. Prior scaling studies compare pre-training and fine-tuning behavior, including sparse models (Tay et al., 2022). Recent work links router shifts to instability during MoE reinforcement learning (Zhang et al., 2026), while counterfactual route replacement tests whether a frozen router selects useful experts on reasoning tokens (Yoon et al., 2026). We instead compare organizational resolutions along a controlled SFT trajectory. Other MoE analyses perturb or identify experts to study safety-relevant behavior (Lai et al., 2025). Our controlled trajectory is narrower: it describes routing during one reasoning-SFT run and does not identify a universal post-training law or a causal source of behavioral change. Our contribution-space intervention is likewise a secondary functional assay, not an expert-importance method. It scales an already selected expert's post-gating contribution and tests whether a baseline margin derivative predicts a particular downstream routing transition. Fisher geometry (Amari, 1998) motivates the local normalization, but the paper's primary object remains observed routing organization rather than a global importance score.

3 Core measurement framework

We distinguish three levels throughout. *Aggregate marginals* summarize expert use with N_{eff}, Gini, or activation frequency. *Local organization* describes one token and layer. *Cross-state comparisons* quantify how two aligned local organizations differ.

For router scores $r_{\ell,t,i}$, let $A_{\ell,t}$ be the selected top-k expert set and let $B_{\ell,t}$ be the fixed five-rank window spanning ranks 6–10, used as a near-cutoff diagnostic throughout the reported analyses. The local safety margin is

$$\gamma_{\ell,t} = r_{\ell,t,(k)} - r_{\ell,t,(k+1)}. \tag{1}$$

For cross-state comparisons at aligned sites from checkpoints or rendered contexts a and b, we report top-k Jaccard $J(A^a, A^b)$, rank-window Jaccard $J(B^a, B^b)$, the Spearman correlation of score ranks for expert IDs shared by the retained top lists, and the geometric margin ratio γ^b/γ^a. These are distinct observables: the scalar margin measures only the rank-k/rank-$(k{+}1)$ gap, while rank and set metrics measure broader score ordering and identity. Cross-family score values are never treated as calibrated on a common numerical scale. For top-8 Qwen and Gemma, this window spans ranks on both sides of the rank-8/rank-9 boundary; for top-6 Kanana and Nemotron, it contains rank 6 and four ranks outside the selected set. It is therefore not a symmetric boundary neighborhood in every architecture, and cross-family results are interpreted as complementary rank-position diagnostics rather than directly equivalent boundary-neighborhood measurements. We use the tuple

$$\mathcal{C}(x,\ell,t) = \big(A(x,\ell,t), B(x,\ell,t), r_B(x,\ell,t), \gamma(x,\ell,t)\big) \tag{2}$$

as shorthand for local competition organization. The paper compares $\mathcal{C}^{\text{Base}}(x,\ell,t)$ with post-trained states at aligned contexts and sites; it does not assume that one global expert map $E_i^{\text{Base}} \mapsto E_j^{\text{Post}}$ is sufficient. Here r_B denotes scores restricted to the rank window; full retained score profiles are used only as supporting representations.

3.1 Supporting functional assay

For a selected expert contribution $u_i = g_i(h) f_i(h)$, we apply a temporary post-gating scale $u_i \mapsto (1+\delta_i)u_i$ and allow all downstream routers to execute naturally. The source site (ℓ_s, t_s) is the selected contribution at source layer ℓ_s and the token containing the question's last non-punctuation content character. A target routing site (ℓ_d, t_d) may be any monitored later-layer position from t_s through the final prompt token; the next-token distribution is evaluated only at that final prompt position. For the target site's rank-k/rank-$(k+1)$ margin γ_{ℓ_d,t_d}, we differentiate the baseline margin with respect to this scale and define

$$g_{i\to\ell_d,t_d} = \left.\frac{\partial \gamma_{\ell_d,t_d}}{\partial \delta_i}\right|_0, \qquad \delta_i^* = -\epsilon\,\text{sign}(g_{i\to\ell_d,t_d}), \qquad \mathcal{R}_{i\to\ell_d,t_d} \tag{3}$$

We choose the source-perturbation sign to decrease the positive baseline margin to first order, so $\gamma_{\ell_d,t_d}(\delta_i^*) \approx \gamma_{\ell_d,t_d}(0) - \epsilon|g_{i\to\ell_d,t_d}|$. Thus $\mathcal{R}$ measures predicted margin erosion relative to the available gap, and $\mathcal{R} \geq 1$ predicts first-order boundary crossing. The intervention changes neither parameters nor the source router's selected IDs. $\mathcal{R}$ is a baseline-only vulnerability assay for a specified downstream boundary, not a definition of reorganization or expert importance. We prospectively compare the highest-risk source with matched-low

and random active sources at the same destination; separately, on records that actually flip, we restore baseline expert IDs and measure the recovered next-token KL. Full algebra, numerical audits, selection rules, and negative response-geometry results are in Appendix C.

4 Experimental protocol

Models and execution. The controlled trajectory starts from Qwen3-30B-A3B-Base and is the primary experiment. Cross-sectional triangulation adds Qwen3-30B-A3B and Qwen3-30B-A3B-Thinking-2507. Qwen has 48 MoE layers, 128 experts, and top-8 routing. Cross-architecture score audits add Kanana-2, Nemotron-3-Nano, and Gemma 4 Base/post-trained conditions using each model's native ranking score and top-k. Expert IDs are compared only within families. The functional assay uses one middle source layer and all later MoE routers; family-specific layer, shared-expert, rendering, and numerical details appear in the appendix. Intervention models run in FP32 with FP64 KL evaluation.

Data and target protocol. We use GSM8K (Cobbe et al., 2021), pinned to a recorded dataset revision. A seeded permutation of the official GSM8K training split produces 512 build prefixes and 128 disjoint analysis prefixes after filtering to 31–127 tokens. These prompts are held out from the SFT corpus, not sampled from the official GSM8K test split. Answers are excluded from primary manifests. The prompt asks the model to solve the given problem and provide the answer. For functional interventions, the source token t_s contains the question's final non-punctuation content character. Candidate target router events (ℓ_d, t_d) span later layers and positions from t_s through the final prompt token. The next-token distribution is evaluated at the final prompt token preceding generation. Downstream routing executes naturally after the source perturbation. The F build uses 512 prefixes; joint interventions use a fixed first 16-prefix held-out subset for the primary pilot. After that analysis was frozen, the same 40 $\epsilon = 0.01$ interventions were re-evaluated on all 128 held-out prefixes, producing 5,120 prefix records. This expanded run is a robustness and heterogeneity analysis, not an independent confirmation, and it does not access the sealed redesign confirmation split.

Two prospective G4-local splits contain 128 new prefixes each. The first is disjoint from all historical Stage 2 and reconstructed redesign IDs; the second also excludes the first G4-local split. Selection uses neither answers nor modified outputs. Cross-checkpoint and cross-architecture comparisons use the first prospective split, recomputing active experts, derivatives, destinations, and controls independently for every checkpoint/condition. Each uses its own 512-prefix marginal atlas. Gemma IT OFF and ON share weights but not input token sequences: OFF ends with an empty thought block, whereas ON adds a system turn containing `<|think|>`. We therefore treat them as two rendered contexts, not as an isolated causal intervention on thinking. For the cross-family score audit, we reuse the first 64 fixed prompt strings. Each checkpoint is tokenized independently and records its native top-list score profile; Gemma uses pre-top-k FP32 softmax probabilities and Nemotron uses $\sigma(q) + b$. Scores are compared by normalized prompt position, not by forced token identity. No modified forward is run for these replay artifacts. For trajectory analysis, Qwen Thinking greedily generates up to 512 tokens on 64 prompts. We store its generated token IDs, replay the complete prompt plus continuation, and record top-8 identities at every layer. Common-Trace Replay then teacher-forces each identical stored token sequence through Qwen Base, A3B, and Thinking, recording top-8 IDs, normalized weights, the rank-9 outside expert, and the rank-6:10 window. The score replay additionally retains the top-16 router IDs and their sorted scores at every layer and token; all 64 token-sequence hashes match across checkpoints. Base and A3B do not naturally generate these traces; their measurements are conditional routing responses to Thinking-generated content.

For the controlled training trajectory, we full-parameter fine-tune Qwen3-30B-A3B-Base on a frozen 32,000-example reasoning-SFT corpus. We fix the optimizer, cosine 10^{-5}–10^{-6} schedule with 5% warmup, global batch 32, data order, seed, maximum length 8,192, and router auxiliary coefficient .001. We analyze 15 checkpoints from the same continuous run: every 50 steps through step 500 and every 100 steps thereafter through step 1,000; full model, optimizer, scheduler, and RNG state are restored at steps 200, 500, and 800. At every checkpoint and at initialization, we teacher-force the same 64 stored traces (36,853 prompt-plus-continuation tokens) and retain top-16 IDs and scores, selected top-8 IDs, and normalized

weights. Exact and normalized problem-overlap checks against the SFT corpus are zero for both these traces and the frozen 512-prompt atlas.

The behavior audit reuses the first G4-local split's 128 prompts. It is therefore distinct from the earlier 128-prefix Stage-2 analysis set (zero shared IDs), but is not an independent prompt sample from the first prospective functional study. The prompts come from the official GSM8K training split and are held out from the SFT corpus, with zero exact and normalized corpus overlap. Each state uses greedy decoding, seed 20261006, and at most 512 new tokens; correctness compares the last generated numeric value with the reference. We report exact-match accuracy, generated-token length, and cap rate. Base through step 900 are complete and pass record-count and unique-ID checks. Step 1,000 is unfinished and is not included in any behavioral trend claim.

Metrics and uncertainty. Trajectory summaries use medians across the 48 layers. All traces are divided into length-normalized early, middle, and late thirds of the observed generated continuation. The answer segment is reported only for traces containing the explicit closing thought delimiter. Temporal baselines compare top-8 Jaccard at lags 1, 2, 4, and 8 with random position pairs drawn within the same layer and trajectory; $8/(256-8) = .0323$ is shown only as a uniform-set approximation. CommonTrace agreement uses top-8 Jaccard, top-1 match, and rank-window Jaccard. Exact rank-8/rank-9 agreement is retained as a supplementary high-cardinality diagnostic. For the controlled training trajectory, baseline-relative drift at step s is $D_s^{\mathrm{top8}} = 1 - J(A^0, A^s)$, $D_s^{\mathrm{win}} = 1 - J(B^0, B^s)$, and $D_s^{\mathrm{rank}} = 1 - \rho_s(0, s)$. Sampled checkpoints are 50 steps apart through step 500 and 100 steps apart thereafter. Adjacent-checkpoint drift replaces initialization by the preceding checkpoint; when comparing unequal intervals we normalize it as $\widetilde{V}_s = V_s/(s - s^-)$. We report the fraction of step-1,000 endpoint change already present at step 50 and the sum of adjacent-checkpoint dissimilarities relative to endpoint dissimilarity. This operational sum is descriptive; we do not claim that every component, in particular $1 - \rho_s$, defines a metric-space path length. We also report layer-median N_{eff}, Gini, and the geometric margin ratio. Uncertainty uses 20,000 paired bootstrap resamples of the 64 traces; tokens are not treated as independent. Top-8 Jaccard, five-member rank-window Jaccard, and rank correlation have different scales, so we compare temporal profiles within each observable without claiming cross-metric intrinsic sensitivity. The prospective functional studies use paired prefixes with high-risk, matched-low, and random active sources at one fixed destination. Intervals use 20,000 prefix-bootstrap draws and exact McNemar tests use paired transitions. Cross-checkpoint and mode contrasts remain observational. The permutation control learns each layer's assignment on one prefix half and evaluates the other. Complete intervention grids, finite-difference amendments, numerical gates, and protocol lineage are reported in the appendix.

5 Results

5.1 The controlled run reveals multi-timescale reorganization

The cross-sectional checkpoints below cannot reveal when routing organization forms. We therefore first summarize the primary within-run result. Under fixed optimization and identical replay content, effective expert count increases from 66.60 to 89.76 and Gini falls from .601 to .456 by step 1,000. Baseline-relative top-8, rank-window, and shared-ID rank drift reach .453, .763, and .321, while the geometric cutoff-margin ratio contracts to .636.

Step 50 already accounts for 76.6% [75.7,77.5], 85.0% [84.4,85.6], and 79.6% [78.2,81.0] of the respective endpoint local drifts, but only 28.2% [25.5,31.1] and 26.7% [24.3,29.4] of the effective-count and Gini changes. Margin contraction is intermediate at 50.5% [48.7,52.2]. Adjacent-state top-8/rank-window/rank drift is .347/.649/.256 for $0 \to 50$ and, after normalizing the wider final interval, .054/.120/.031 per 50 steps for $900 \to 1000$. Observed drift therefore decelerates but persists; it does not establish convergence; individual intervals exhibit small fluctuations. The summed adjacent-checkpoint dissimilarities are 2.458/5.327/1.525, substantially larger than the corresponding endpoint dissimilarities .453/.763/.321, showing repeated local turnover.

Because step 50 is the earliest saved checkpoint, we cannot localize its change within the first 50 updates. Layer-by-step maps show broad depth involvement rather than a narrow layer sequence (Appendix N). The behavior audit through step 900 is also non-monotonic in accuracy while mean length and the 512-token cap rate increase. This is evi-

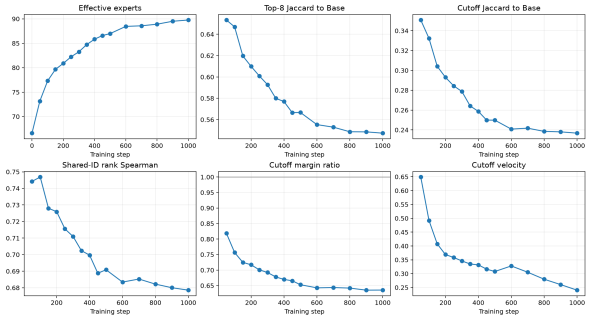


Figure 2: Primary controlled trajectory under identical replay content. Local organization undergoes most net endpoint drift by step 50, aggregate allocation changes more gradually, and adjacent-state turnover persists through step 1,000.

dence against treating routing drift as a scalar capability proxy, not evidence of causal independence. Full checkpoint values appear in Appendix N.

5.2 Cross-checkpoint and cross-architecture evidence

The controlled trajectory's separation of resolutions is also visible in cross-sectional Qwen states. Across 512 prompts, Base, A3B, and Thinking have similar median effective expert counts (50.0, 51.9, 49.3) and context specialization, yet exact top-8 preservation relative to Base is only 13.5% and 17.6%. Thus coarse utilization can conceal token-level reorganization.

Common-Trace Replay removes generated-text differences by teacher-forcing the same 64 Thinking-generated sequences through all three states. From early to late positions, top-8 Jaccard with Base decreases from .654 to .631 for A3B and from .623 to .581 for Thinking; rank-window agreement is lower still. Score replay separates this identity turnover from a scalar-margin account: Thinking's shared-ID rank agreement declines from .727 to .687 while its margin ratio remains near 1.04. These are conditional responses to fixed Thinking content, not natural Base or A3B generations.

Two cross-fitted global-relabeling controls test whether changing expert names explains the low overlap. Functional-fingerprint matching improves held-out expert-profile cosine but reduces route overlap; matching optimized for token routing is 88% identity and changes held-out top-8 Jaccard by only .00011. The tested global permutations therefore do not recover local correspondence.

Kanana, Nemotron, and Gemma4 fixed-prompt audits reproduce the empirical separability of rank, margin, rank-window membership, and selected identity, but not a common magnitude or positional direction. Because they use native scores, architecture-specific tokenizers, and normalized-position matching, they provide external triangulation rather than replications of the controlled temporal law. Full tables, figures, bootstrap intervals, and implementation qualifications are in Appendix B.

5.3 A functional check: local routing transitions have consequences

Organization describes which competition is present; risk describes how easily that competition can be crossed. We retain only two main-text claims: a baseline-only margin-erosion score prospectively enriches a prespecified target transition, and restoring changed expert IDs suppresses most next-token divergence among records that actually transition. All retrospective calibration, component, non-max, cross-family, and interaction diagnostics are supporting analyses in the appendix.

Prospective targeted transitions. G4-local tests the vulnerability of a selected local competition directly. For each new prefix, all selection features are baseline-only; the high, matched-low, and random conditions share the exact destination event and margin. The selection manifest is hashed before modified forwards. No matched-low selection is relaxed.

On the first independent split, high-risk sources flip the targeted event in 59/128 prefixes (0.4609), versus 13/128 (0.1016) for matched-low and 15/128 (0.1172) for random. The primary paired high-minus-low effect is 0.3594 with 95% prefix-bootstrap CI [0.2812,0.4453]; all 46 discordant pairs are high-only (exact McNemar $p = 2.84 \times 10^{-14}$). On a second disjoint split, rates are 46/128 (0.3594), 12/128 (0.0938), and 16/128 (0.1250), with primary effect 0.2656 [0.1875,0.3438] and 34 high-only discordant pairs ($p = 1.16 \times 10^{-10}$). Both independently frozen gates pass. The secondary fixed-effect summary is 0.3086 [0.2523,0.3649], but $I^2 = 62.2\%$; we therefore emphasize study-specific estimates.

These are prospective targeted validations, not randomized treatment effects: source conditions are selected from baseline geometry. Nevertheless, matching fixes the destination and balances source marginals, selection precedes outcomes, and the exact preselected event is primary. The evidence therefore establishes predictive intervention util-

ity of the relational score as a vulnerability assay, rather than treating risk as a definition or proxy for identity reorganization.

Routing restoration. We next analyze all 225 old E5-A records with an observed transition. Restoring all changed expert IDs suppresses a median 0.99677 of natural next-token KL; 96.9% of records suppress more than half and 88.4% suppress more than 90%. Restoring IDs and baseline weights yields a similar median (0.99661). Restoring only the first transition has the same median suppression; its KL equals the all-flip clamp exactly in 216/225 cases. Nine exceptions, including two negative first-clamp restoration fractions, show that later transitions can interact nonadditively. Zero-intervention clamps are exactly inert and natural flip counts are reproduced in 225/225 cases. Routing restoration therefore accounts for most divergence in observed transition records. Because the analysis conditions on flips and is retrospective, it is not a population causal-effect estimate.

The supporting component study finds that $|g|/\gamma$, unlike either component alone, induces the selected transition; a non-max study places all 12 observed transitions in the highest frozen-risk quintile. Cross-family enrichment is heterogeneous, and a balanced 54-site pilot observes only two transitions and does not resolve an interaction between reorganization and vulnerability. These results keep the functional claim narrow: routing transitions are predictable and consequential in the tested setting, but routing reorganization itself is not a vulnerability score. Within-condition risk ordering is observed across the tested families, but Nemotron's multi-step finite-difference audit fails; we therefore make no cross-step claim about its derivative magnitude or mechanism. Full tables and numerical audits are in the appendix.

6 Discussion

What changes? Routing change is multi-resolution and exhibits distinct empirical temporal profiles. We use multi-timescale descriptively: we do not fit dynamical time constants or assume exponential convergence. In the controlled run, local identity, rank-window, and rank measures acquire roughly three quarters or more of their net step-1,000 drift by step 50, while effective count and Gini acquire less than one third of their endpoint change. Later checkpoints add relatively little net displacement but continue to turn over locally, as shown by the large sum of adjacent-checkpoint dissimilarities relative to endpoint dissimilarity. In the cross-sectional Qwen comparison, coarse usage is instead similar even though aligned top-8 sets and rank-6:10 windows differ. Same-content replay makes both findings conditional routing comparisons rather than consequences of different generated text. The decomposition is measurable in four families, but its direction and magnitude are not universal.

What is the right object? The evidence favors the context- and site-conditioned tuple $\mathcal{C}(x, \ell, t)$ over the tested global summaries. Aggregate utilization alone misses token-level turnover, while the tested global expert relabelings do not recover held-out routing correspondence. Expert ID continuity therefore contains signal, but these global summaries are insufficient to describe local organization. The preregistered stationary-graph stability tests and their criteria are reported in the appendix rather than used as a separate headline. This is an empirical resolution claim, not an exhaustive impossibility result over all global summaries or a proof that measured score changes causally produce identity turnover.

What does it mean functionally? Organization and vulnerability answer different questions: $\mathcal{C}$ describes which competition is present, whereas $\mathcal{R}$ describes how easily a specified source perturbation crosses it. Prospective targeting, component ablation, and non-max ordering support $\mathcal{R}$ as a transition assay; routing restoration connects observed expert replacement to output divergence. The small $O^B \times \mathcal{R}$ pilot does not establish an interaction, so reorganization is not itself a vulnerability score. Several families show within-condition enrichment, but operating distributions and checkpoint/mode contrasts are heterogeneous, and Nemotron's derivative audit is unstable across finite- difference steps. All training-state comparisons are observational; Gemma mode additionally changes rendered context.

What does behavior add? Through step 900, exact-match accuracy on prompts held out from the SFT corpus fluctuates rather than improving monotonically as routing drift accumulates, while output length and censoring increase markedly. This is evidence against treating routing reorganization as a scalar capability proxy. It is not evidence that routing and behavior are unrelated: the audit is small,

heavily censored at later checkpoints, and lacks the unfinished step-1,000 endpoint. The defensible conclusion is temporal non-equivalence, not causal independence.

7 Conclusion

Post-training-associated MoE change is multi-resolution. Under shared content, local score, rank, rank-window, and selected-set organization can change even when cross-sectional coarse usage remains similar. Along one controlled reasoning-SFT run, however, aggregate load also redistributes: expert use becomes more diffuse while local identity and rank-window organization turn over. Their timescales differ: most net local endpoint drift is already present at step 50, whereas aggregate allocation continues to evolve through step 1,000. Persistent adjacent-state turnover and low endpoint-to-summed-dissimilarity ratios show that a nearly flat baseline-relative curve is not the same as a stationary router. The completed behavior audit through step 900 shows fluctuating rather than monotonically improving exact-match accuracy as routing drift accumulates and also exposes increasing length and censoring; routing reorganization is therefore not a scalar capability proxy. The tested global summaries do not recover this structure; in particular, functional correspondence does not imply routing correspondence, and the routing-optimal expert permutation is nearly identity with negligible held-out gain. Across four families, the decomposition is measurable but its direction and magnitude are heterogeneous. Functional relevance is distinct rather than automatic: normalized margin erosion prospectively targets routing transitions, and flip-conditioned restoration links expert replacement to output divergence, while reorganization itself is not a vulnerability score. Causal training attribution, a complete uncensored behavioral endpoint, cross-trajectory generalization, task and layer transfer, and trajectory-level functional consequences remain open.

Limitations

The controlled evidence is one Qwen reasoning-SFT trajectory, one seed, and one task family. It supports a within-run temporal description, not a universal post-training law or causal attribution of behavioral change to routing. Step 50 is the earliest saved intermediate checkpoint, so substantial change observed there cannot be localized to particular earlier updates. Checkpoint intervals also widen after step 500; we normalize adjacent drift when comparing intervals and do not claim convergence or a fitted dynamical time constant.

The behavior audit on prompts held out from the SFT corpus is complete only through step 900, and more than half of later generations reach the 512-token cap. It therefore does not identify a final accuracy endpoint or establish independence between routing and behavior. It reuses the first G4-local prompt split, so the behavioral and functional analyses are not independent at the prompt-sample level, although they measure different outcomes. This split is disjoint from the earlier Stage-2 analysis set. Common-Trace Replay fixes content but conditions Base and A3B on Thinking-generated sequences they need not produce naturally. Cross-family comparisons are weaker normalized-position analyses with architecture-specific scores and tokenizers; numerical levels are not directly comparable.

The global-relabeling result applies only to the tested cross-fitted Hungarian controls. It does not exclude context-dependent correspondences or changes in experts' full functions. Functional interventions act at one source layer and evaluate downstream router events at potentially different token positions and the immediate next-token distribution at the final prompt position. Prospective sources are selected from baseline geometry rather than randomized, and routing restoration conditions on observed flips. Nemotron's multi-step derivative audit fails, so its within-model enrichment does not support a derivative-shift mechanism. The small reorganization–risk pilot does not establish absence of an interaction. Further tasks, seeds, training trajectories, layers, and uncensored behavior remain open.

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A Disclosure and reproducibility statements

Writing assistance. Generative AI tools assisted experimental code development, debugging, workflow documentation, analysis scripts, figure and table generation, and drafting and editing this manuscript. The authors reviewed AI-assisted outputs against saved configs and artifacts, executed numerical consistency checks, and retained negative results. No generative AI system is an author. The human authors take responsibility for the final content, claims, code, and artifacts.

Reproducibility. Supplementary code and artifacts specify checkpoint, pinned dataset revision, prompt and target-token protocol, split hashes, intervention definition, precision, seeds, and acceptance checks. One script regenerates every figure and result table from saved records. Appendices report derivative protocol lineage, negative gates, prospective selection hashes, routing-clamp validation, prefix-bootstrap trend intervals, and estimator uncertainty. The controlled trajectory manifest additionally records all 15 checkpoint paths and hashes, resume lineage, fixed trace hashes, and the exact/normalized SFT-overlap audit; all route streams pass content and score-order validation. Behavior states through step 900 pass record-count and unique-ID checks, while step 1,000 is explicitly marked incomplete and excluded. The reserved confirmation set is excluded.

B Cross-checkpoint and cross-architecture evidence

B.1 Cross-sectional routing provides external triangulation

We record the selected expert IDs and normalized top-8 weights at every question token, every layer, and all 512 prefixes for Qwen Base, A3B, and Thinking: 35,368 prompt tokens and 13.58 million route assignments per checkpoint. Figure 3 shows three levels of organization. First, layer-wise effective expert count is nearly unchanged: its median is 50.0, 51.9, and 49.3. Second, unsupervised eight-cluster context specialization is also stable, with median normalized $I(E;C)$ of .01270, .01258, and .01255. Thus the atlas does not support the hypothesis that Thinking globally uses fewer experts or becomes more specialized on these prompt tokens.

Reorganization instead appears in identity and competition. Versus Base, the median layer-wise top-16 usage Jaccard is .778 for both A3B and Thinking, and top-1 expert identity is preserved on 64.7% and 69.4% of question tokens. The full top-8 set is preserved on only 13.5% and 17.6%, while the exact rank-8/rank-9 boundary pair is preserved on only 2.6% and 3.6%. Population edge sets are more stable but show the same ordering: median top-1000 Jaccard is .798/.794 for co-activation, .769/.779 for adjacent-token transitions, and only .532/.545 for boundary-pair competition. Post-trained states therefore preserve coarse load and much of the co-activation/transition skeleton while substantially reorganizing token-level membership and especially the marginal competition. A3B is not uniformly closer to Base than Thinking, so the atlas does not support a simple monotone training progression.

Generated-trajectory context. We greedily generate 64 Qwen Thinking continuations and use their stored token sequences for the Common-Trace experiment below. Descriptively, routing remains dynamic: median effective expert count increases from 60.8 to 68.2 from early to late observed positions, and lag-8 top-8 Jaccard (.156) remains above a matched within-trajectory shuffle (.132). Because most traces hit the 512-token cap and only Thinking is generated naturally, the full trajectory analysis is supporting rather than a checkpoint-level claim (Appendix).

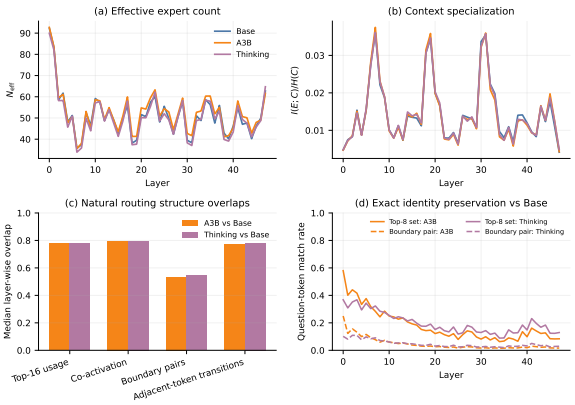


Figure 3: Natural-routing atlas over 512 GSM8K prompts. (a) Effective expert count and (b) checkpoint-independent context specialization are nearly unchanged. (c) Layer-wise population edge overlap remains highest for co-activation and adjacent-token transitions and lowest for boundary pairs. (d) Exact question-token expert identity rapidly diverges with depth, with the rank-8/rank-9 competing pair least preserved.

B.2 Same-content replay isolates checkpoint-associated organization

Natural generation confounds checkpoint identity with generated text. We remove that content mismatch by teacher-forcing the same 64 Thinking-generated token sequences through Base, A3B, and Thinking. This is a routing-response comparison under shared content, not a claim that Base or A3B would naturally produce those sequences. Substantial checkpoint differences remain. Relative to Base, median-across-layer top-8 Jaccard for A3B is .654, .643, and .631 in early, middle, and late generation; Thinking is less aligned at .623, .603, and .581. Its decline from early to late is also larger (.042 versus .023). Top-1 match remains around .61–.65 and does not show a monotone stage effect (Table 5).

The rank window shows particularly low agreement. Rank-6:10-window Jaccard changes from .334 to .322 for A3B and from .270 to .252 for Thinking. Exact rank-8/rank-9 pair agreement is only 1–2%, but we treat that as supplementary because exact pairs have high cardinality. Length-normalized curves corroborate the stage summaries: Thinking remains less aligned than A3B and reaches .566 top-8 Jaccard in the final bin, versus .616 for A3B. Therefore, holding prompt and continuation tokens fixed, checkpoint identity is associated with routing reorganization, and the Thinking state differs more from Base than ordi-

Table 1: Score-level Common-Trace decomposition with Base. Values are medians across layers for the indicated generation segment. γ ratio is the geometric mean across site-level ratios; score-rank agreement uses shared expert IDs in the top-16 lists.

Checkpoint	Segment	γ ratio	Score-profile distance	Sha
A3B	Early	0.995	.172	
	Middle	1.056	.170	
	Late	1.059	.173	
Thinking	Early	1.040	.179	
	Middle	1.047	.175	
	Late	1.043	.177	

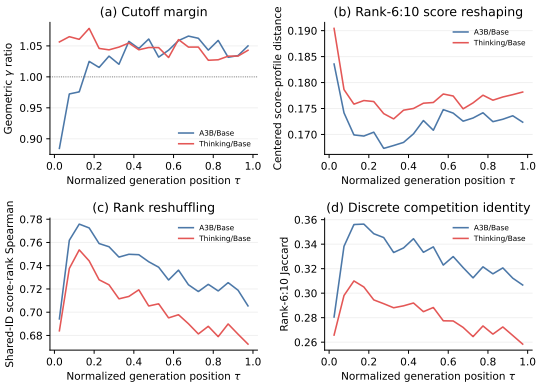


Figure 4: Score-level decomposition under Common-Trace Replay. (a) Thinking shows a modest, position-stable cutoff-margin expansion relative to Base, while A3B is less uniform. (b) Centered rank-6:10 score profiles are reshaped in both comparisons. (c) Spearman agreement of scores for expert IDs shared by the two top-16 lists declines with position, more strongly for Thinking. (d) The same positions exhibit discrete rank-window turnover. Rank-slot Spearman is excluded because both score vectors are sorted by construction.

nary A3B on these traces, especially at later observed positions.

B.3 Score-level decomposition of rank-window reorganization

Identity turnover alone does not say whether the router scores moved or merely reshuffled their local order. We therefore rerun the same-content replay while retaining the top-16 router scores (the existing ID-only replay is unchanged). For each checkpoint we report the top-8/top-9 margin γ, the centered and scale-normalized score-profile distance at ranks 6–10, and Spearman agreement of score values for expert IDs appearing in both top-16 lists. The latter is computed on shared expert IDs; Spearman on the already sorted rank slots would be tautologically one and is omitted.

Figure 4 gives the position-dependent decomposition. Thinking has geometric-mean cutoff-margin ratios of 1.040, 1.047, and 1.043 relative to Base in early, middle, and late generation, respectively. A3B has 0.995, 1.056, and 1.059, so its margin shift is less uniform. At the same time, Thinking's shared-ID score-rank Spearman falls from .727 to .704 to .687, and its rank-6:10 Jaccard falls from .287 to .282 to .272. The corresponding A3B values are .753, .742, .727 and .334, .335, .322. Thus Thinking's stronger late discrete turnover coexists with a modest margin expansion and progressive local rank reshuffling; the identity result is not explained by a single scalar margin shift. Raw score levels are interpreted only as same-architecture, same-content relative quantities because router-score parameterizations need not be globally calibrated across checkpoints.

The third edge of the Qwen checkpoint triangle is available without another forward pass. Direct Thinking–A3B agreement also declines from early to late: top-8 Jaccard falls from .691 to .643, rank-window Jaccard from .384 to .335, and shared-ID score-rank Spearman from .792 to .748. In contrast, the geometric cutoff-margin ratio changes only from 1.067 to .980. Thus the result is not only two separate distances from Base: A3B and Thinking themselves become less aligned over the observed continuation even though their scalar cutoff margins remain similar.

B.4 A global expert permutation does not explain turnover

Low same-ID overlap could arise if post-training merely permuted a stable set of expert roles. We test this alternative with two layer-wise Hungarian assignments, each learned on 32 prefixes and evaluated on the disjoint half. Functional- fingerprint matching improves held-out routing-profile cosine for Thinking–Base from .829 to .901, yet top-8 and rank-window Jaccard fall from .613 to .366 and .298 to .163. Thus expert-level routing correspondence is not token-level route correspondence. More directly, a second assignment optimizes binary top-8 activation-vector Jaccard on the training tokens. It is 88% fixed points, gains only about 5×10^{-4} over identity on its training objective, and changes held-out top-8 Jaccard by only .00011. The routing-optimal global permutation is therefore nearly identity and provides no substantive recovery. Figure 5 shows the primary Thinking–Base control; complete pairs, intervals,

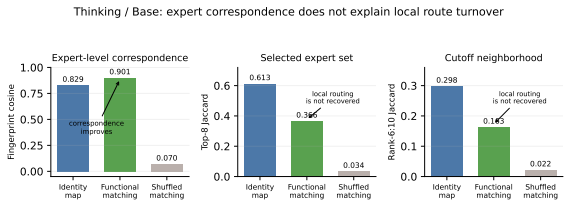


Figure 5: Three-level Qwen Thinking/Base permutation control. Hungarian assignments are learned from expert routing fingerprints on one prefix half and evaluated on the other. Functional matching raises held-out fingerprint cosine (.829 to .901), confirming expert-level correspondence, but lowers top-8 and rank-window overlap (.613 to .366 and .298 to .163). Shuffled matching shows the random-permutation expectation. Thus correspondence improves without recovering local routing. Separately, the routing-optimized map is nearly identity (88% fixed points) and changes held-out top-8 Jaccard by only .00011.

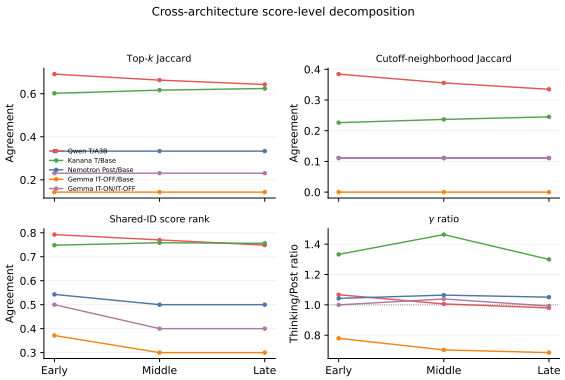


Figure 6: Cross-architecture score-level decomposition on fixed prompt strings. Qwen and Kanana use the generated/common or plain replay described above; Nemotron and Gemma use normalized-position matching with checkpoint-specific tokenizers. Across families, score-rank reshaping and rank-window identity are measurable separately, while margin magnitude and positional profiles are heterogeneous.

and objectives are in the appendix. These routing-derived fingerprints are not evidence of semantic or full input–output equivalence.

B.5 Cross-architecture score-level decomposition

Kanana. We next repeat only the four most diagnostic measurements in an independent MoE architecture. We teacher-force the same 64 plain GSM8K prompt strings through Kanana Base and Thinking, using their native tokenizer and actual choice score $r_i = \sigma(q_i) + b_i$. Kanana has top-6 routing, so γ is the rank-6/rank-7 gap; ranks 6–10 form the fixed five-rank diagnostic window, with only rank 6 on the selected side. This replay contains 4,409 prompt tokens across 47 MoE layers and performs no intervention.

Table 3 shows that Kanana also has low rank-window identity agreement (.226–.245) despite higher top-6 set overlap (.602–.624), together with shared-ID score-rank agreement around .75. Its margin ratio is larger and nonmonotone (1.333, 1.463, and 1.300), and its identity/rank metrics do not exhibit Qwen's late-position decline. The independent family therefore reproduces the continuous-score/discrete-identity decomposition but not a universal positional direction. We interpret this as architecture- and checkpoint-dependent heterogeneity, not a failed replication.

Nemotron and Gemma4. To test whether the decomposition was specific to the two preceding families, we repeat the same fixed-prompt normalized-position analysis for the existing Nemotron Base–Post checkpoints and for Gemma4 Base, IT-OFF, and IT-ON. Nemotron uses its corrected choice score $\sigma(q) + b$ and top-6 routing; Gemma uses the FP32 pre-top-8 softmax probability. Each comparison contains 64 identical prompt strings, but checkpoint-specific tokenizers, so these are normalized position comparisons rather than token-identity common traces.

The results in Table 2 and Figure 6 reinforce the heterogeneous interpretation. Nemotron Post/Base has top-6 Jaccard .384, .348, and .326 and rank-window Jaccard .146, .132, and .123 from early to late, while its margin ratio remains close to one (1.043, 1.064, 1.051). Gemma IT-OFF/Base has very low local identity agreement (top-8 Jaccard .143–.148 and rank-window Jaccard about .055) and a margin ratio that falls from .779 to .685. Holding Gemma IT weights fixed, IT-ON/IT-OFF has margin ratios 1.000, 1.039, and .992; its score-rank and rank-window identity profiles shift, but neither follows the Qwen late-decline pattern. Thus the robust cross-family statement is that continuous score profiles and discrete rank-window identities can be analyzed as separate quantities; their amplitude and positional direction depend on architecture, checkpoint, and rendered context.

The positional contrasts are not inferred from point estimates alone. A prefix-bootstrap over the 64 replay prefixes gives the early-minus-late intervals in Table 4. For Qwen, all three identity/rank contrasts exclude zero. Kanana has a small early-to-late increase in overlap for the two identity mea-

Table 2: Nemotron and Gemma4 score-level replication. Entries are medians over router layers for early/middle/late normalized positions. Top-k is 6 for Nemotron and 8 for Gemma; rank-6:10 is the fixed five-ID diagnostic window. It is not symmetric around the top-6 boundary.

Pair	Segment	γ ratio	Shared-ID ρ_s	Window Jaccard	Top-k Jaccard
Nemotron Post/Base	Early	1.043	.543	.146	.384
	Middle	1.064	.500	.132	.348
	Late	1.051	.500	.123	.326
Gemma IT-OFF/Base	Early	.779	.371	.056	.143
	Middle	.703	.300	.056	.143
	Late	.685	.300	.054	.143
Gemma IT-ON/IT-OFF	Early	1.000	.500	.111	.231
	Middle	1.039	.400	.111	.231
	Late	.992	.400	.111	.231

Table 3: Direct Qwen Thinking–A3B comparison and minimal Kanana replication. Values are medians across layers. γ ratios are Thinking/A3B for Qwen and Thinking/Base for Kanana; top-k is 8 and 6, respectively.

Pair	Segment	γ ratio	Shared-ID ρ_s	Rank-window Jaccard	Top-k Jaccard
Qwen Think./A3B	Early	1.067	.792	.384	.691
	Middle	1.006	.770	.356	.663
	Late	.980	.748	.335	.643
Kanana Think./Base	Early	1.333	.748	.226	.602
	Middle	1.463	.758	.237	.617
	Late	1.300	.756	.245	.624

Table 4: Prefix-bootstrap early-minus-late contrasts over 64 replay prefixes. Values are differences in the layer-median stage summaries; intervals contain 95% bootstrap replicates (20,000 draws). Positive values indicate larger early agreement.

Pair	Metric	Early − Late	95%
Qwen Think./A3B	Top-k Jaccard	.047	[.038,.05
	Rank-window Jaccard	.048	[.039,.05
	Shared-ID ρ_s	.044	[.036,.05
Kanana Think./Base	Top-k Jaccard	−.023	[−.034, −.01
	Rank-window Jaccard	−.016	[−.027, −.00
	Shared-ID ρ_s	−.004	[−.012, .00

Table 5: Common-Trace Replay agreement with Qwen Base. Values are medians across 48 layers under identical Thinking-generated token sequences.

Checkpoint	Segment	Top-8 Jaccard	Rank-window Jaccard	Top-1 ma
A3B	Early	.654	.334	
	Middle	.643	.335	
	Late	.631	.322	
	Answer*	.618	.307	
Thinking	Early	.623	.270	
	Middle	.603	.263	
	Late	.581	.252	
	Answer*	.576	.246	

sures, while its score-rank agreement and margin ratio are nonmonotone; the data therefore do not support a common monotone positional trajectory.

Supporting global-summary tests. Earlier experiments tested whether a single full response matrix or stationary source-to-router graph could replace the local description. In a 16-prefix pilot, full and diagonal response predictors were similar ($R^2 = .837$ and .841); over 128 prefixes, pooled full-matrix R^2 fell to $-.088$. A preregistered static graph also failed edge-rank, top-edge, and out-strength stability gates. These negative results motivate local measurement but are not a second headline; full geometry, estimator, and graph diagnostics appear in the appendix.

C Supporting response geometry

For prefix x, let $p_0(y \mid x)$ and $p_\delta(y \mid x)$ be baseline and intervened next-token distributions. We define

$$s_i(x,y) = \left.\frac{\partial \log p_\delta(y \mid x)}{\partial \delta_i}\right|_{\delta=0}, \qquad F = \mathbb{E}_{x,y\sim p_0(\cdot|x)}[s(x,y)s(x,y)^\top], \tag{4}$$

and accumulate $\widehat{F} = N^{-1}\sum_n s_n s_n^\top$ in FP64. Within a fixed-routing region, $\mathbb{E}_x D_{\mathrm{KL}}(p_0 \| p_\delta) = \frac{1}{2}\delta^\top F \delta + o(\|\delta\|_2^2)$. We compare the full predictor with its diagonal counterpart. The small pilot supports local quadratic prediction, but the full matrix does not beat the diagonal and pooled absolute magnitude fails the expanded-prefix robustness check. We therefore treat this geometry as supporting history rather than the paper's main object.

D Natural identity turnover

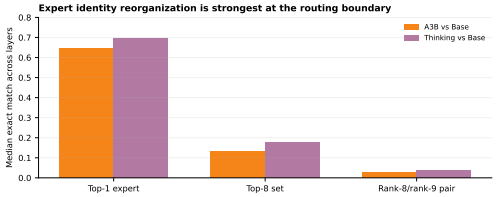


Figure 7: Median exact identity preservation across layers. The top-1 expert remains relatively stable, the full top-8 set usually changes, and the rank-8/rank-9 competing pair changes on more than 96% of natural question-token events.

E Generated-trajectory descriptive figure

F Implementation validation

Layer 47 was used for the formal finite-difference gate to minimize accumulated downstream round-off; the same adapter and algebra are used at layer

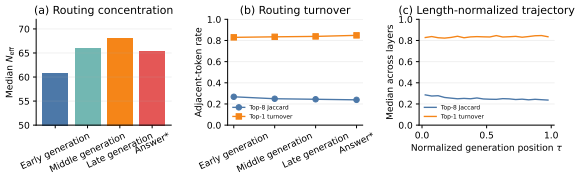


Figure 8: Routing along 64 Qwen Thinking generated trajectories. Effective expert count increases and adjacent-token top-8 overlap decreases from early to late observed generation. Most generations hit the 512-token cap; answer estimates use only delimiter-positive traces.

Table 6: Numerical and algebraic validation checks; all listed checks pass.

Check	Setting	Result
Zero intervention	layer 23, FP32	max logit difference 0
Inactive experts	layer 47, 20 cases	autograd and FD max 0
Finite difference	layer 47, 80 active	median error 0.735%
Finite difference	layer 47, 80 active	P90 error 4.942%
Finite difference	layer 47, 80 active	sign agreement 100%
Repeat score	fixed seed/prefix	max difference 0
F symmetry	$N = 512$	relative error 0
Streaming vs. offline	$N = 512$	1.57×10^{-16}
Positive semidefinite	$N = 512$	$\lambda_{\min} = -2.24 \times 10^{-20}$

23. An initial BF16 score path was rejected because its numerical discrepancy was too large; all reported response and intervention results use strict FP32.

G Detailed local prediction results

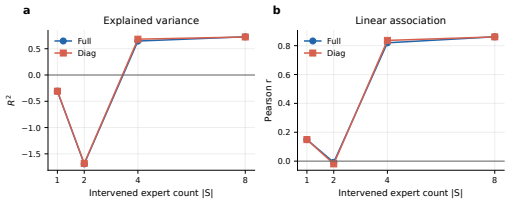


Figure 9: Descriptive local metrics by intervention subset size. Each stratum contains 10 interventions; the pooled result is primary.

Negative R^2 for small subsets reflects tiny observed KL and only 10 cases per stratum. Full and diagonal are identical for $|S| = 1$ by construction.

Table 7: Local prediction stratified by intervened expert count.

$\lvert S\rvert$	Predictor	n	R^2	Pearson r	Spearman ρ_s
1	Full	10	−0.308	0.149	0.423
1	Diagonal	10	−0.308	0.149	0.423
2	Full	10	−1.691	−0.008	0.067
2	Diagonal	10	−1.682	−0.019	0.067
4	Full	10	0.646	0.820	0.503
4	Diagonal	10	0.682	0.836	0.503
8	Full	10	0.727	0.861	0.903
8	Diagonal	10	0.724	0.861	0.879

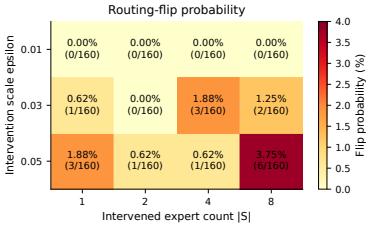


Figure 10: Flip probability by perturbation scale and subset size. Each cell contains 160 records. No record flips at $\epsilon = 0.01$. Percentages are descriptive because records share prefixes and interventions.

H Routing analysis details

I Sampling stability and estimator redesign

J Artifact scope

All figures are generated from saved JSONL, JSON, CSV, and NumPy artifacts. The pilot local prediction unit is an intervention mean over 16 prefixes; the E4 expanded unit is an intervention mean over 128 prefixes; the routing unit is one intervention–prefix record. These units remain separate to avoid inflating sample size. E4 artifacts are stored under `artifacts/followup/e4_expanded_local_v2/` and include the 20,000- replicate bootstrap results. The generation manifest records output hashes and that the redesign confirmation set was not accessed. The Qwen–A3B and Kanana score-replay trend intervals are generated from `artifacts/graph/reorganization_trend_bootstr` using 20,000 prefix resamples; no additional model forward is required.

K Derivative protocol lineage

The original stock sequence-wide G0 full-vector finite-difference gate failed. Component audits localized the discrepancy to the stock in-place expert aggregation: `torch.func.jvp` and low-

Table 8: Record-level routing-transition summary.

Quantity	Estimate	Uncertainty/reference
Records	1,920	16 prefixes × 120 interventions
Records with flip	17	cluster rate CI [0.0021, 0.0203]
Affected interventions	14	of 120
ρ ROC–AUC	0.9977	Mann–Whitney $p = 4.24 \times 10^{-16}$
Median ρ, flip	33.415	no-flip P90 = 0.892
Mean KL, flip	6.822×10^{-4}	no-flip = 2.765×10^{-9}
Mean log-KL contrast	flip minus no-flip	cluster CI [6.77, 9.62]

Table 9: Prefix-split stability; lower difference and higher overlap are better.

Estimator	N	Half difference	Top-8 overlap	Decision
Categorical $M = 4$	512	0.762	0.613	unstable
Categorical RB $M = 4$	128	0.953	0.693	unstable
Hutchinson $M = 8$	128	0.892	0.564	unstable
Hutchinson $M = 4$	512	0.566	0.631	narrow ga

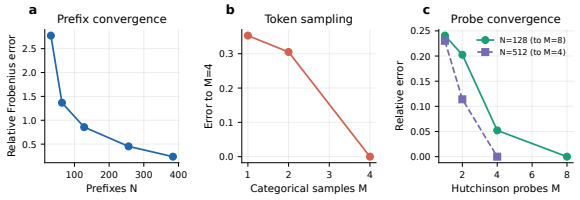


Figure 11: Sampling diagnostics: nested-prefix estimates, categorical token sampling, and Hutchinson probe convergence. Probe convergence does not imply convergence over independent prefixes.

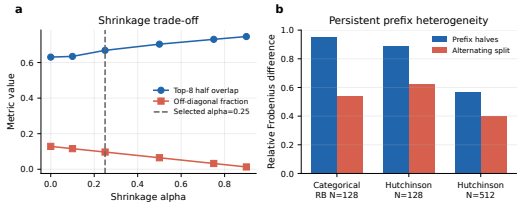


Figure 12: Exploratory redesign. Shrinkage trades subspace stability against off-diagonal magnitude, while prefix heterogeneity persists across estimators. Outcome tuning did not select a superior estimator, so confirmation stayed sealed.

level forward AD agreed with each other but not with reverse mode or stable finite differences. On the documented out-of-place `index_add` amendment, forward AD, low-level dual tensors, `autograd.functional.jvp`, reverse mode, and finite differences agree at the audited router event. Full-vector forward/reverse relative error is 1.67×10^{-6} with cosine effectively one; stable $\epsilon = 10^{-3}$ finite difference has relative error 0.0129 and cosine 0.99992.

A separate 8-prefix primal audit verifies that the amendment changes only expert aggregation, not parameters, routing policy, intervention definition, targets, or thresholds. Small FP32 differences are reported as accumulation-order effects rather than requiring bit identity. The stock failure remains recorded; it is not retroactively relabeled. The amended scalar router-margin derivative subsequently passes its primary AD oracle gate. Position-dependent full-vector FD remains an explicit numerical diagnostic, not a replacement gate.

L Functional assay details

Prospective transition rates. Across the first two independently frozen 128-prefix splits, high-risk, matched-low, and random sources change the exact targeted event at rates .4609/.1016/.1172 and .3594/.0938/.1250. The paired high-minus-low effects are .3594 [.2812,.4453] and .2656 [.1875,.3438]; all 46 and 34 discordant pairs are high-only. These are baseline-selected predictive contrasts rather than randomized treatment effects.

Component and non-max controls. On a disjoint 64-prefix split, sensitivity-only and margin-only selectors each produce 0/64 transitions, while normalized $|g|/\gamma$ produces 32/64 versus 6/64 for a source-matched same-boundary control. On another 64 prefixes, all 12 transitions among 512 non-max candidates fall in the highest frozen-risk quintile (Table 15).

Cross-family summary. High-minus-low effects are .359/.336/.219 for Qwen Base/A3B/Thinking, .172/.148 for Nemotron Base/Post, .125/.250/.203 for Gemma Base/IT-OFF/IT-ON, and .414/.289 for Kanana Base/Thinking. Only the Qwen Thinking–Base effect contrast is resolved; the other checkpoint/mode contrasts are not. These results support transfer of within-condition enrichment, not a universal robustness shift.

Reorganization–risk pilot. The balanced 54-site $O^B \times \mathcal{R}$ pilot observes only two target transitions, both at high risk but at opposite reorganization levels. It does not resolve an interaction between boundary reorganization and vulnerability.

E5-A logs 2,206,080 downstream events for 120 interventions and 128 prefixes, forming 15,360 record aggregates. It finds 551 flip events in 225 records and 68 interventions. No flip satisfies $2\|\Delta q\|_\infty < \gamma$. Among affected records, 145/225 flips occur at the record's minimum-margin event.

Table 10: Exploratory no-flip shrinkage tuning smoke ($n = 11$).

Predictor	R^2	MSE	MAE
Shrinkage ($\alpha = 0.25$)	0.5042	4.1769×10^{-19}	4.5973×10^{-10}
Raw response	0.5057	4.1644×10^{-19}	4.5792×10^{-10}
Diagonal	0.4991	4.2201×10^{-19}	4.6515×10^{-10}

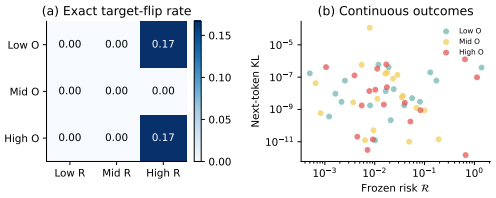


Figure 13: Balanced selected-site mechanism pilot. Exact target-flip rates and next-token KL after crossing rank-window reorganization with frozen Base risk. Only two target flips occur, both at high risk but at opposite reorganization levels; the experiment does not establish an $O^B \times \mathcal{R}$ interaction.

Flip-event median margin is 0.003109 versus 0.091072 for nonflips; median router displacement is 0.03374. Flips concentrate in layers 45–46, followed by 42, 41, and 44. Event maxima reproduce legacy record-level ρ exactly.

Table 11: Routing restoration over all 225 observed transition records.

Clamp	Median suppression	P10	Fraction > 0.5	Fraction > 0.9
First changed event	0.99677	0.85672	0.9600	0.8756
All changed IDs	0.99677	0.86761	0.9689	0.8844
All IDs and weights	0.99661	–		

First- and all-clamp KL are exactly equal in 216 records. The nine exceptions are retained. The minimum all-clamp suppression is -0.0106; one slightly negative case prevents an absolute restoration claim.

M Prospective selection and cross-architecture details

Generated trajectory and Common-Trace Replay audit. The Thinking trajectory set contains 64 unique prompts, 32,344 generated tokens, and 36,853 prompt-plus-continuation tokens. Only 6/64 traces finish before the 512-token cap and 23/64 contain the closing thought delimiter. The replay grids for Base, A3B, and Thinking each contain the same 64 complete sequences and pass record-count, uniqueness, shape, normalized-weight-sum, and content-hash checks. Their record SHA256 values begin `43f83e250209bced`, `ddfcd7ea2520f97e`, and `d2a7c4b9eccc9558`. No sealed confirmation record is read. Exact rank-8/rank-9 agreement with Base is .022/.022/.022 across A3B's early/middle/late segments and .016/.015/.014 for Thinking. We retain these values for completeness but prefer rank-window Jaccard because exact pair identity is a high-cardinality statistic.

Table 12: Held-out functional-fingerprint permutation control. Entries show raw Jaccard, Jaccard after cross-fitted Hungarian relabeling, and matched-minus-raw change with 95% prefix-bootstrap CI (20,000 draws).

Pair	Set	Raw	Matched	Δ (95% CI)
A3B/Base	Top-8	.655	.437	$-.218$ [$-.222$, $-.215$]
	Rank-window	.348	.213	$-.135$ [$-.138$, $-.132$]
Thinking/Base	Top-8	.613	.366	$-.247$ [$-.251$, $-.243$]
	Rank-window	.298	.163	$-.135$ [$-.139$, $-.132$]
Thinking/A3B	Top-8	.672	.442	$-.230$ [$-.234$, $-.225$]
	Rank-window	.366	.218	$-.148$ [$-.152$, $-.145$]

N Controlled training-trajectory details

The training trajectory uses Qwen3-30B-A3B-Base and full-parameter BF16 reasoning SFT. The frozen corpus contains 32,000 examples; maximum serialized length is 8,192. We fix training and data-order seeds to 20261004, disable dataloader shuffling, use global batch 32, a cosine schedule from 10^{-5} to 10^{-6} over 1,000 planned steps with 5% warmup, and router auxiliary coefficient .001. Checkpoints 50–200 belong to the initial process; checkpoints 250–500 continue from checkpoint 200, checkpoints 600–800 continue from checkpoint 500, and checkpoints 900–1000 continue from checkpoint 800. Each resume restores model, optimizer, scheduler, and RNG state. Each checkpoint contains eight distributed shards and its recorded iteration matches its directory name.

All 16 states, including Base, replay the same 64 traces and 36,853 prompt-plus-continuation token IDs. Each route stream passes record count, uniqueness, shape, descending top-16 score, normalized selected-weight sum, and content-hash checks. Neither these 64 problems nor the separate 512-prompt atlas has exact or normalized overlap with the SFT corpus. Figure 14 gives the two layer-by-step views. Both show broad depth involvement; they do not support a simple narrow-band or sequential layer hierarchy.

The adjacent-checkpoint drifts over 50-step intervals through step 500 for top-8/rank-window/rank

Table 13: Controlled trajectory. Drift is one minus agreement with Base; values are medians across 48 layers.

Step	N_{eff}	Gini	Top-8	Rank-6:10 window	Rank	γ_s/γ_0
0	66.60	.601	.000	.000	.000	1.000
50	73.15	.563	.347	.649	.256	.819
100	77.31	.534	.353	.668	.253	.757
200	80.88	.510	.390	.707	.274	.717
300	83.26	.495	.407	.721	.289	.692
400	85.83	.481	.423	.741	.300	.670
500	86.98	.473	.433	.750	.309	.653
600	88.45	.464	.445	.759	.317	.643
800	88.91	.461	.451	.761	.318	.642
1000	89.76	.456	.453	.763	.321	.636

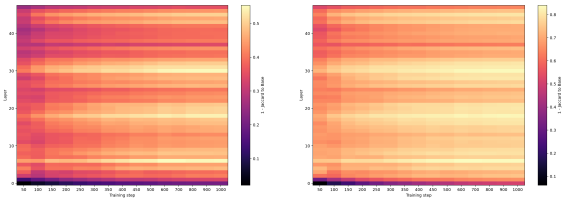


Figure 14: Layer-by-training-step baseline-relative drift. Left: selected top-8 set. Right: rank-6:10 window. The common color interpretation is $1-$Jaccard, but each panel has its own empirical scale.

are: .347/.649/.256 ($0 \rightarrow 50$), .232/.492/.155 ($50 \rightarrow 100$), .187/.408/.114 ($100 \rightarrow 150$), .162/.369/.095 ($150 \rightarrow 200$), .162/.359/.097 ($200 \rightarrow 250$), .156/.346/.093 ($250 \rightarrow 300$), .146/.335/.087 ($300 \rightarrow 350$), .152/.332/.092 ($350 \rightarrow 400$), .140/.316/.082 ($400 \rightarrow 450$), and .139/.308/.080 ($450 \rightarrow 500$). For reference, at step 500 the 95% paired-bootstrap intervals for top-8 Jaccard, rank-window Jaccard, and shared-ID rank Spearman are [.561,.572], [.245,.254], and [.684,.697]; the interval for log margin ratio is [-.437,-.409]. The later 100-step adjacent-checkpoint drifts are .146/.328/.085, .136/.305/.081, .127/.280/.076, .118/.261/.071, and .108/.240/.061 for $500 \rightarrow 600$ through $900 \rightarrow 1000$. Because interval widths differ, this sequence is descriptive rather than a per-update convergence rate. The final triplet is .054/.120/.031 per 50 training steps, so observed drift decelerates but persists and does not establish convergence. At step 1,000 the 95% paired-bootstrap intervals for top-8 Jaccard, rank-window Jaccard, and shared-ID rank Spearman are [.541,.554], [.231,.240], and [.673,.685]; the interval for log margin ratio is [-.463,-.437]. Tokens contribute within a trace but are not bootstrap units.

Step 50 accounts for 76.6% [75.7,77.5], 85.0% [84.4,85.6], and 79.6% [78.2,81.0] of the final top-8, rank-window, and rank drift, compared with 50.5% [48.7,52.2] of margin contraction, 28.2% [25.5,31.1] of effective-count change, and 26.7% [24.3,29.4] of Gini change. These are within-observable fractions and are not a common-scale comparison. Summing adjacent dissimilarities gives 2.458, 5.327, and 1.525, while the corresponding endpoint dissimilarities are .453, .763, and .321. This is an operational sum over saved checkpoints, not a claim that all three quantities define geometric distances.

Held-out behavior audit. The behavior manifest freezes the first G4-local split's 128 prompts (disjoint from the Stage-2 analysis set and zero-overlap with the SFT corpus), greedy decoding, 512 new tokens, seed 20261006, and last-number correctness parser for every state. The completed results are:

Table 14: Held-out behavior along the controlled trajectory. Cap is the fraction of generations reaching 512 new tokens. Step 1,000 is intentionally omitted because its audit is unfinished.

Step	Correct/128	Accuracy	Mean tokens	Cap
0	85	.664	59.1	.008
50	75	.586	151.2	.164
100	73	.570	96.4	.102
150	72	.562	158.1	.172
200	84	.656	308.5	.359
250	69	.539	341.8	.453
300	79	.617	328.1	.398
350	66	.516	366.5	.523
400	79	.617	335.3	.398
450	65	.508	374.2	.547
500	66	.516	375.1	.555
600	63	.492	368.1	.547
700	68	.531	393.7	.594
800	62	.484	406.8	.672
900	68	.531	408.9	.688

Every included state contains 128 unique records and a passing summary. Because later outputs are frequently right-censored at the generation cap, accuracy and length should be read jointly. The missing step-1,000 output directory contains only its frozen configuration; no value is imputed and no partial result enters the manuscript.

P2 reorganization–risk pilot audit. The baseline phase evaluates 1,536 candidate sites (64 traces, one stratified generation position, and 24 downstream layers), then freezes 54 interventions with six in each $O^B \times \mathcal{R}$ cell. Reorganization

cut points are .571 and .750; log-risk cut points are -2.064 and -1.506. Selection and outcome SHA256 values begin `0abeb788ec5e116e` and `679609fbb464bad8`. All selection-hash, active-source, baseline-repeat, target-route, finite-value, and confirmation-isolation checks pass. The 54 sites span 40 prefixes; prefix reuse across different cells is retained in the cluster bootstrap.

P3 score replay audit. The score replay stores 36,853 prompt-plus-continuation tokens for each of Base, A3B, and Thinking, with 48 layers, top-16 IDs, and top-16 FP32 router scores. Each score stream passes record-count, uniqueness, shape, descending-score, normalized-weight, content-hash, and confirmation-isolation checks. The score record SHA256 values begin `7fff8f83b37db70b`, `860f492e19ec66b0`, and `589a29b9d0004042`; the analysis summary is stored under `artifacts/graph/common_trace_score_decomposition_n64/`. The permutation-control summary is stored under `artifacts/graph/expert_permutation_control_n64/`. Its Hungarian map is fit separately for every layer on 32 alternating prefixes and evaluated on the other 32, then the folds are reversed. Fingerprint channels are early/middle/late top-16 score, top-16 presence, top-8 activation, and selected gate weight. All reported Jaccard changes use held-out prefixes and 20,000 prefix-bootstrap draws. The route-optimized control is stored under `artifacts/graph/route_optimized_permutation_qwen_n64/`. It uses the same folds and learns each layer's assignment from binary top-8 activation vectors over 16,126 or 16,218 training-context token sites; no modified-forward outcome enters either assignment. For Thinking–Base, the two folds improve the mean training activation-Jaccard objective by .00047 and .00052 over identity; their non-fixed fractions are .123 and .121.

Full SHA256 values are stored in the released manifests. For compact audit, their leading 16 hexadecimal characters are: first G4 prefix/selection/outcome `e45109de55c77ec`, `287d825c913ed2ff`, and `be8aa423eee30525`; second G4 prefix/selection/outcome `114a664f7a1131e4`, `490d7ba687b3e948`, and `650783637ac75daf`. A3B selection/outcome begin `bb615e3a3f6b8ab5`/`2478bcbdde1836e2`; Thinking begins `1dc57f6e46fdb8f5`/`6377ee7825d3be1f`. Corrected-score Nemotron Base selection/outcome begin `7b72886c99438242`/`c3bfe808238a4ba9`; Nemotron Post begins `e1fc1b80a63d2626`/`fb89b604a1164e44`. All baseline repeats, counts, target boundaries, finite values, and selection hash checks pass. No confirmation data are used.

P0 prospective freeze. The new P0 prefix, component-selection, component-outcome, non-max-selection, and non-max-outcome SHA256 values begin `58969e9184dc3819`, `f80c92e39385f701`, `54b9ad4bff0117a5`, `7d87e7a7763aff6d`, and `e0a44cef3471313f`. The two 64-prefix subsets are disjoint from both earlier prospective G4 splits. Both outcome programs verify the frozen selection hash before loading the model and never recompute selection features.

Table 15: Prospective non-max ordering conditional on near-boundary candidates. One downstream event is sampled per prefix before outcomes; all eight active source directions are retained. Intervals are prefix-cluster bootstrap 95% CIs.

Risk quintile	n	Frozen $\mathcal{R}$ range	Exact target flip
Q1	103	$[2.0\times10^{-6}, 7.8\times10^{-4}]$	.000 [.000,.000]
Q2	103	$[8.2\times10^{-4}, 2.9\times10^{-3}]$	.000 [.000,.000]
Q3	102	$[2.9\times10^{-3}, 8.1\times10^{-3}]$	.000 [.000,.000]
Q4	102	$[8.1\times10^{-3}, 4.0\times10^{-2}]$	.000 [.000,.000]
Q5	102	$[4.0\times10^{-2}, 3.20]$	.118 [.024,.219]

Table 16: Cross-checkpoint selected-event geometry and local susceptibility.

Checkpoint	Median γ	Median $\lvert g\rvert$	Median $\lvert g\rvert/\gamma$	Random flip
Base	0.00229	0.0787	30.44	0.1172
A3B	0.00174	0.0664	29.00	0.1250
Thinking-2507	0.00305	0.0547	15.91	0.0703

For 128 paired Base–Thinking prefixes, selected-event γ is larger for Thinking in 73 cases (57.0%) and smaller in 55; the paired median ratio is 1.44. The median derivative over all eight active sources is smaller in 72 cases (56.2%) and larger in 56; its paired median ratio is 0.83. Hence the aggregate margin shift is statistically resolved but heterogeneous, not pointwise dominance. The stored basis contains all eight source directions only at the selected target event; it cannot recover the unconditional distribution over every nonselected eligible event.

Common-site boundary turnover. Freezing the Base-selected physical (ℓ, t) does not freeze the underlying decision boundary. Qwen Thinking preserves the Base rank-boundary expert pair in only 1/128 prefixes, and Kanana Thinking in only 2/128. At those physical sites, the state-specific Thinking/Base geometric-mean ratios are 40.5 [27.4,60.2] and 52.0 [35.9,75.4] for γ, 0.34 [0.28,0.41] and 0.29 [0.23,0.38] for maximum active-source $|g|$, and 0.0083 [0.0053,0.0129] and 0.0056 [0.0036,0.0087] for maximum $|g|/\gamma$. Because the boundary expert pair changes in nearly every prefix, these are state-specific physical-site operating-point contrasts, not same-boundary geometric effects.

Table 17: Qwen source-layer utilization and functional identity across the checkpoint progression. All utilization statistics use the same 512-prefix build set; source-match rates use the 128-prefix prospective split.

State	Entropy	N_{eff}	Top-16 Jaccard vs. Base	High-risk source match
Base	3.455	31.7	1.000	1.000
A3B	3.490	32.8	.882	.266
Thinking	3.445	31.3	.684	.328

Nemotron amendment and superseded path. Nemotron routes by $r_i = \sigma(q_i) + b_i$, not by raw logits. We therefore define its rank-6/rank-7 margin in corrected score space. An initial raw-logit-boundary analysis is retained only as a superseded diagnostic and is not reported as a scientific result. The corrected protocol independently freezes all 128 target events before modified forwards, perturbs only routed experts (not the shared expert), and uses a central finite-difference derivative with step 10^{-3}. Both corrected 384-record outcome grids pass selection-hash, uniqueness, finite-value, exact final-logit repeat, and exact target-boundary checks. Maximum absolute selection-to-outcome margin drift is 3.81×10^{-6} for Base and 8.39×10^{-5} for Post, while boundary expert identities remain exact.

Nemotron high-risk flip rates over ascending within-model $\mathcal{R}$ quintiles are $[0, 0, 0.038, 0.32, 0.84]$ for Base and $[0, 0, 0, 0.28, 0.84]$ for Post. These bins are diagnostic within-model calibration, not a shared probability scale. Their sharp ordering complements the cross-family functional summary: the risk primitive transfers even though the Post/Base effect and geometry contrasts are unresolved. A multi-step FD audit at eight frozen events fails for both Nemotron states: relative to 10^{-3}, median relative L2 errors are 1.00 at 3×10^{-4} and 0.66–0.71 at 3×10^{-3}, with low sign agreement. The discrete derivatives are FP32 cancellation-limited. This negative numerical result does not alter the already frozen 10^{-3} prospective enrichment, but it rules out strong cross-step claims about Nemotron's derivative magnitude or checkpoint shift.

Kanana architecture and derivative-associated variation. Kanana-2 uses a DeepSeek-V3-style MLA/MoE implementation with 48 decoder layers: layer 0 is dense and layers 1–47 are MoE. Each MoE layer has 128 routed experts, top-6 selection, and one shared MLP whose width equals two routed experts. We use physical source layer 23 and destinations 24–47, intervene only on routed contributions, and leave the shared MLP untouched. Expert identities are selected by $\sigma(q) + b$; contribution weights instead gather uncorrected sigmoid scores, renormalize over top-6, and multiply by 2.448. This explicitly separates ranking score from contribution weight. Stock/out-of-place aggregation relative L2 is 6.18×10^{-8} for Base and 6.04×10^{-8} for Thinking.

Base and Thinking selection/outcome hashes begin `e3d6e497867da070`/`df86a9ca7703c59e` and `3951599cfa5e1f25`/`6da1b7ab6dc96a71`. Each checkpoint uses a distinct 512-prefix atlas and a pre-outcome frozen selection. Both 384-record grids pass exact inactive/zero-delta, baseline repeat, boundary, finite-value, uniqueness, and selection-hash checks, with no relaxed matches. Ascending-risk flip rates are $[0, .038, 1, 1, .92]$ and $[0, 0, .231, .92, .96]$. The nonmonotonic top-bin fluctuation is compatible with 25–26 prefixes per bin; the defensible claim is sharp low-to-high stratification, not deterministic calibration.

Both checkpoints use the identical plain prompt rather than the Thinking chat template. Consequently the comparison is cleaner with respect to input context, but remains an observational weight-state association and does not evaluate generated reasoning. A 12-event follow-up varies the central-FD step over 3×10^{-4}, 10^{-3}, and 3×10^{-3}. Relative to 10^{-3}, median relative L2 errors are 0.055/0.010 for Base and 0.021/0.008 for Thinking; sign agreement and active-source Spearman are 1.0 throughout. The derivative stability gate therefore passes, closing the step-size concern while leaving

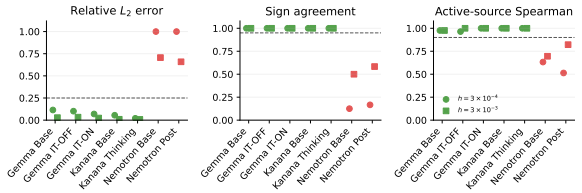


Figure 15: Multi-step finite-difference audit relative to the frozen $h = 10^{-3}$ oracle. Gemma and Kanana conditions pass the preregistered relative error, sign, and rank gates at 3×10^{-4} and 3×10^{-3}; both Nemotron states fail. Green/red indicate the overall condition-level decision; dashed lines are acceptance thresholds.

the selected-event comparison limitation intact.

Gemma architecture and mode-conditioned context. The executable Gemma 4 26B-A4B model has 30 MoE text layers, 128 routed experts, top-8 selection, and no shared expert. Its router first computes FP32 softmax probabilities, selects top-8 identities, then renormalizes selected weights and applies per-expert scales. We therefore define γ as the rank-8/rank-9 gap in the pre-selection softmax probabilities. Stock in-place and derivative-safe out-of-place aggregation differ by relative L2 7.09×10^{-8}. All three 512-prefix atlases and 384-record outcome grids pass inactive, zero-delta, repeat, boundary, finite-value, and selection-hash checks; no matched-low selection is relaxed.

Eight-event multi-step FD audits also pass for Base, IT-OFF, and IT-ON. Relative to the frozen 10^{-3} step, median relative L2 errors at 3×10^{-4} are 0.115, 0.102, and 0.070, and at 3×10^{-3} are 0.028, 0.034, and 0.024; sign agreement is 1.0 throughout and median active-source Spearman is at least 0.964. Thus all reported Gemma conditions pass the same numerical stability gate.

Table 18: Multi-step derivative audit. Each pair reports the median at $h = 3 \times 10^{-4}$ / 3×10^{-3} relative to the frozen 10^{-3} estimate. Spearman and sign agreement are over active source directions.

Condition	Relative L_2	Sign agreement	Spearman	Decision
Gemma Base	.115/.028	1.00/1.00	.976/.976	Pass
Gemma IT-OFF	.102/.034	1.00/1.00	.964/1.00	Pass
Gemma IT-ON	.070/.024	1.00/1.00	1.00/1.00	Pass
Kanana Base	.055/.010	1.00/1.00	1.00/1.00	Pass
Kanana Thinking	.021/.008	1.00/1.00	1.00/1.00	Pass
Nemotron Base	1.00/.706	.125/.500	.632/.696	Fail
Nemotron Post	1.00/.659	.167/.583	.514/.822	Fail

Gemma Base, IT-OFF, and IT-ON high-risk flip rates over ascending $\mathcal{R}$ quintiles are $[0, 0, 0, 0, 0.76]$, $[0, 0, 0.115, 0.92, 0.96]$, and $[0, 0, 0, 0.36, 0.96]$. OFF and ON use identical IT weights but distinct official chat renderings. In an audited prefix they contain 68 and 71 tokens and share no terminal suffix: OFF appends an empty thought block, whereas ON prepends a system turn with `<|think|>`. Accordingly, the resolved ON/OFF geometry shift is evidence for mode-conditioned context dependence, not an isolated causal effect of thinking.